

Solution.

Forward pass:

$$h_1 = \sigma(x_1) = \frac{1}{1 + e^{-x_1}}, \quad h_2 = \sigma(x_2) = \frac{1}{1 + e^{-x_2}}$$

$$o_{\min} = \min(h_1, h_2), \quad o_{\max} = \max(h_1, h_2)$$

$$y = o_{\min} + o_{\max}$$

Backpropagation:

$$\frac{\partial E}{\partial x_i} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h_i} \cdot \frac{\partial h_i}{\partial x_i}, \quad i = 1, 2$$

Sigmoid derivative:

$$\frac{\partial h_i}{\partial x_i} = \sigma(x_i)(1 - \sigma(x_i))$$

Since $\min(h_1, h_2) + \max(h_1, h_2) = h_1 + h_2$, we have $y = h_1 + h_2$, thus:

$$\frac{\partial y}{\partial h_1} = 1, \quad \frac{\partial y}{\partial h_2} = 1$$

Therefore:

$$\frac{\partial E}{\partial x_1} = \frac{\partial E}{\partial y} \cdot \sigma(x_1)(1 - \sigma(x_1))$$

$$\frac{\partial E}{\partial x_2} = \frac{\partial E}{\partial y} \cdot \sigma(x_2)(1 - \sigma(x_2))$$

Answer:

$$\boxed{\frac{\partial E}{\partial x_1} = \frac{\partial E}{\partial y} \cdot \sigma(x_1)(1 - \sigma(x_1))}, \quad \boxed{\frac{\partial E}{\partial x_2} = \frac{\partial E}{\partial y} \cdot \sigma(x_2)(1 - \sigma(x_2))}$$